AME 323: Project 2: Wind Tunnel Design

Spring 2026

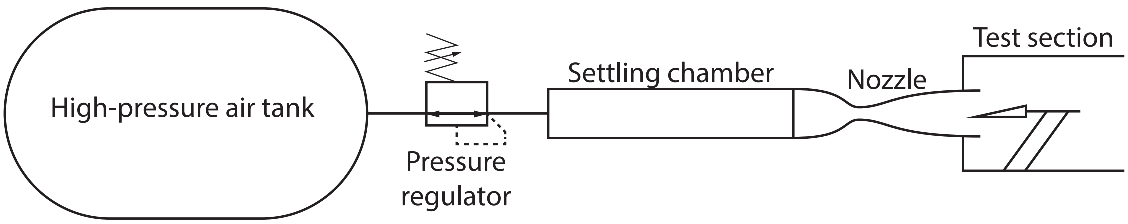

Figure 1: Schematic of a blowdown wind tunnel.

1 Introduction

You are tasked with producing the rough design of a Mach 6 supersonic blowdown wind tunnel with a planar 2D nozzle. This will consist of several related tasks:

- production of a nozzle contour design code using the method of characteristics;
- design of a nozzle contour and characterization of its flow field;
- estimation of the aerodynamic forces on the nozzle;
- estimation of tunnel run envelope; and
- a short technical report detailing your aerodynamic design.

2 Tasks

2.1 Task 1: Nozzle design code

Design a computer code employing the method of characteristics to calculate the wall contour of the expansion section of a 2-D, uniform flow, supersonic nozzle with an exit Mach number of 6. Use the following parameters in the design of your nozzle:

- Assume the throat is a circular arc with radius of curvature equal to the throat half-height (such a shape can help avoid boundary-layer separation). Note: Do not compute a minimum-length nozzle with sharp corners at the throat as shown in several textbooks or the Matlab File Exchange.
- Start by approximating the expansion through your throat region with N equally-spaced Prandtl-Meyer waves. Start with small N for simplicity but design your code to take any value of N. Note: You cannot start exactly at the throat since $\mu = 90°$ at that point.
- Grading note: Make sure your program runs on any machine with the appropriate language installed (i.e., make sure you also submit any programs/scripts upon which your code depends). Any work required in order to make it run while grading will cause points to be deducted.

2.2 Task 2: Nozzle contour

- Perform a grid convergence study on the nozzle design problem using your code. In other words, identify a suitable known parameter to be used to measure the error in your calculations and vary N to determine the dependence of the error on how many initial waves are used. Make a plot of the error as a function of N.

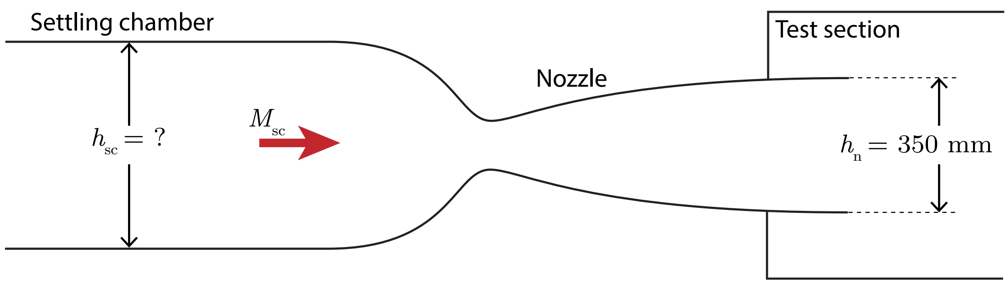

Figure 2: Close-up view of the settling chamber, nozzle, and test section.

- Plot the nozzle expansion contour and the contour map of Mach number inside the nozzle from throat to exit. Make sure you have used enough N for a reasonably accurate nozzle contour. Justify what constitutes a suitable accuracy.
- Plot the centerline distributions of M, p/p_0, and T/T_0 along the centerline of the nozzle from the throat to the exit. Determine the pressure ratio required to operate the nozzle with no shocks or waves at the exit (i.e. P_{cr3}).
- Plot the Mach number, pressure, and temperature contours inside the nozzle expansion.
- Determine the length of the triangle of uniform flow that extends inside the nozzle.

2.3 Task 3: Estimate tunnel characteristics

Use your design to estimate the run characteristics for your tunnel. Figure 1 shows a schematic of the tunnel system including a pressure tank, variable pressure regulator, and the tunnel. Assume that the nozzle exhausts as a free jet into the test section, which is held at constant vacuum level by a vacuum system not shown. A closer view of the tunnel is shown in Figure 2.

Minimum total temperature: In order to avoid liquefaction, the free-stream temperature must be greater than 55 K. Determine the minimum total temperature required to operate the tunnel without liquefaction.

Operating pressure and mass flow range: A regulator supplies a constant pressure that can range from 0 psia to 500 psia to your tunnel system. Suppose your vacuum system is capable of maintaining a back pressure of $p_b = 0.1$ psia regardless of the tunnel settings.

1. What is the minimum total pressure required for steady operation of the tunnel with no shocks at the exit?
2. If your nozzle exit has a height of 350 mm and the entire nozzle has a width of 500 mm, what is the mass flow rate across your operating pressure range? Plot the relationship.
3. You have a pre-existing compressed air system available to provide air for your tunnel. The tank is 40 m^3 in volume and air is stored at 300 K and held approximately constant by a thermal matrix in the tanks. Determine how long your tunnel can run as a function of total pressure. Plot the relationship.
4. Given the range of total pressure and minimum total temperature previously calculated, determine the free-stream unit Reynolds number as a function of total pressure. Plot the relationship. Note: Unit Reynolds number is the Reynolds number without a length scale, $Re' = \rho U/\mu$ and has units of 1/m. Viscosity can be estimated using Sutherland's law of viscosity (Equation 1).

$$\mu = \mu_{\text{ref}} \left(\frac{T}{T_{\text{ref}}}\right)^{3/2} \left(\frac{T_{\text{ref}} + S}{T + S}\right) \tag{1a}$$

$$\mu_{\text{ref}} = 1.716 \times 10^{-5} \text{ Pa} \cdot \text{s}, \quad T_{\text{ref}} = 273.15 \text{ K}, \quad S = 110.4 \text{ K} \tag{1b}$$

Additional geometric characteristics: How much space do you need?

1. What is your nozzle expansion length?
2. What cross-sectional area is required in your settling chamber such that $M \leq 0.05$? (A settling chamber is the section just upstream of the converging section that often contains flow conditioning devices.) Assume that its width is also 500 mm and height is h_{sc}.

Nozzle forces: Again assuming a width of 500 mm and an exit height of 350 mm, estimate the forces on the nozzle. This should include the thrust due to the air accelerating through the nozzle, including the settling chamber, as well as the vertical force due to pressure on a single nozzle plane (you can ignore the contraction section for this and only calculate it on the expansion region you designed). Use the same h_{sc} calculated previously. Note: You do not need to produce a mechanical design for the nozzle.

2.4 Task 4: Report

Prepare a design report (maximum 10 pages of text, 20 pages including figures) detailing your code, nozzle, and tunnel. The report should adhere to the following guidelines:

- Include in-text citations and a references section containing any sources you use to produce your report other than the course notes.
- Explain the basic operation of your design code.
- Include and discuss all plots requested in Task 2.
- Include and discuss all calculations and plots requested in Task 3.
- Add any additional plots where deemed appropriate.
- You are free to do all calculations using a computer, but the process for calculating each requested quantity should be explained using equations and discussion of algorithms
- Perform a grid convergence study on the nozzle design problem using your code. In other words, identify a suitable known parameter to be used to measure the error in your calculations and vary N to determine the dependence of the error on how many initial waves are used. Make a plot of the error as a function of N. as appropriate.
- Acknowledge and explain any use of AI/LLMs in an appendix.
- Do not exceed 10 pages.